

CRM & Business Development System

Complete Module Documentation

Implementation • Pages • APIs • Architecture • Dashboard Integration • Git Workflow • QA Results

Document Item	Details
Module	Customer Relationship Management (CRM) & Business Development (BDM)
Django applications	<code>src/apps/crm/</code> & <code>src/apps/bdm/</code>
Backend	Django + Django REST Framework
Database	Supabase PostgreSQL
Email Infrastructure	Brevo SMTP (smtp-relay.brevo.com)
Background processing	Celery / Redis
QA environment	Development; PostgreSQL 15 / Supabase, Django 4.2, DRF
QA date	18 August 2026
Document status	Implementation and QA documentation

Prepared as an implementation handover / knowledge-transfer document: This document provides complete verification of the CRM and BDM backend applications, public forms, lead lifecycle management, automated client account creation, Brevo SMTP integration, and QA proof of evidence.

1. Executive Overview

The Aurexion CRM & BDM module is the platform's commercial management and business development system. Its purpose is to manage the complete sales lifecycle from receiving public inquiries (Contact Forms, RFP Engine, Project Estimator, Quote Requests), ingesting them into the Business Development Manager (BDM) desk, assigning approved leads to Sales Executives, scheduling client meetings with automated email notifications, and auto-provisioning Client Portal user accounts upon deal closure.

The module is intentionally focused on client relationship management rather than being a generic accounting package. The approved scope contains four functional areas: Public Inbound Portals, BDM Intake & Assignment Console, Sales Executive CRM Desk, and Client Portal Account Management.

In simple terms: Client submits inquiry form → BDM reviews & assigns lead to Sales Executive → Sales Executive schedules meeting & emails client → Sales Executive marks deal Won/Lost → System auto-creates Client Account (password: client@2026) and dispatches welcome credentials via Brevo SMTP.

1.1 What problem does the module solve?

- Provides a centralized place for BDM and Sales staff to manage client inquiries.
- Provides a public experience for prospective clients to submit Contact Forms, RFPs, and Estimator details.
- Provides a controlled intake process with validation and secure PostgreSQL storage.
- Creates a unique tracking reference (AUR-LEAD-XXXX) for every successful submission.
- Provides Sales Executives with a meeting scheduling modal that dispatches automated emails via Brevo SMTP.
- Protects client data and internal notes through backend authorization and RBAC.
- Provides automated client account creation upon marking deals WON (default password: client@2026).

1.2 Scope boundaries

The baseline CRM/BDM module does not include generic accounting functionality such as payroll, ledger management, inventory management, automated telephony dialers, or third-party ERP integration.

2. CRM & BDM Module Structure and Pages

For documentation purposes, the module is organized into six core functional pages/areas. The page count below refers to the six approved user-facing functional areas.

#	Functional Page / Area	Primary User	Main Purpose
1	Public Inbound Portals	Public Visitor / Client	Submit project inquiries, RFP details, estimator scope, or contact messages.
2	BDM Dashboard & Inbound Desk	BDM / Super Admin	Review total inbound leads, inspect submission details, decline lost leads, or assign to Sales Executives.
3	BDM Contact Forms Page	BDM / Super Admin	Dedicated page filtering strictly contact_form submissions with search, source filtering, and assignment modal.
4	Sales Executive CRM Console	Sales Executive	View assigned leads funnel, active follow-ups, win rate metrics, and sales activity ledger.
5	Sales Executive Contact Forms Desk	Sales Executive	Displays approved & assigned contact forms with direct Schedule Meeting, Mark Won, and Mark Lost controls.
6	Lead Detail & Meeting Action Desk	Sales Executive / BDM	Detailed lead workspace for scheduling meetings, sending client emails, recording notes, and completing stage transitions.

2.1 Page 1 — Public Inbound Portals

This is the public-facing contact and inquiry area used by clients to submit project inquiries.

- Name — client full name.
- Email — validated client email address.
- Phone — client phone number.
- Company & Industry — business information.
- Source — contact_form, rfp_form, estimator, request_quote.
- Description & Subject — project requirement details.
- Auto-generated unique reference ID (e.g. AUR-LEAD-XXXX).

2.2 Page 2 — BDM Dashboard & Inbound Desk

This is the operational intake area used by Business Development Managers to triage inbound leads.

- Key metrics: Total Leads, Assigned Leads, Unassigned Leads, Conversion Rate.
- Inbound Form Submissions Desk with Decline and Assign actions.
- Modal for assigning leads to active Sales Executives.

2.3 Page 3 — BDM Contact Forms Page

Dedicated BDM management page for contact forms.

- Filters strictly contact_form submissions.
- Real-time search by name, email, company, or reference ID.

2.4 Page 4 — Sales Executive CRM Console

Operational console for Sales Executives to track assigned opportunities.

- Performance metrics: Total Assigned Leads, Qualified Leads, Won Deals, Win Rate %.
- Approved & Assigned Leads desk with Schedule Meeting, Mark Won, and Mark Lost controls.
- Schedule Meeting modal with automated client email dispatch via Brevo SMTP.

2.5 Page 5 — Sales Executive Contact Forms Desk

Displays approved & assigned contact forms filtered strictly for active sales actions.

- Schedule Meeting button opening date/time, type, link, and notes input.
- Mark Won button triggering automatic Client User creation (password: client@2026).
- Mark Lost button capturing mandatory decline reason.

2.6 Page 6 — Lead Detail & Meeting Action Desk

Detailed workspace for an individual lead.

- Chronological activity timeline, follow-up history, and internal notes.
- Direct lifecycle stage transition controls.

2.7 Approved Lead Lifecycle Stages

Stage	Meaning
NEW	Inquiry has been successfully submitted and received.
UNDER_REVIEW	Inquiry is being evaluated by BDM.
CONTACTED	Meeting scheduled or initial client touchpoint completed.
QUALIFIED	Lead verified as active commercial opportunity.
PROPOSAL_SUBMITTED	Proposal or scope quotation delivered to client.
NEGOTIATION	Final terms, contract terms, or pricing adjustments in progress.
WON	Deal closed. System auto-creates Client User account and emails credentials.
LOST	Lead declined or lost. Reason recorded in audit log.

3. End-to-End CRM & BDM Workflow

1. Client fills out a Contact Form, RFP, or Estimator on the website.
2. Public Lead API validates payload and generates unique reference (AUR-LEAD-XXXX).
3. Lead is saved to Supabase PostgreSQL with source='contact_form' and status='new'.
4. BDM opens the BDM Dashboard (/bdm/dashboard) or Contact Forms page (/bdm/contact-forms).
5. BDM inspects submission details, client contact info, and inquiry brief.
6. BDM can decline invalid leads (status='lost') or assign valid leads to a Sales Executive.
7. Approved & assigned lead appears on Sales Executive Dashboard (/crm/dashboard).
8. Sales Executive clicks Schedule Meeting on the assigned lead card.
9. Sales Executive enters Date & Time, Meeting Type, Meeting Link, and Agenda Notes.
10. Sales Executive clicks Send Meeting Email & Schedule.
11. System creates LeadFollowUp record, updates status to 'contacted', and dispatches email via Brevo SMTP.
12. Sales Executive conducts client meeting and negotiates project terms.
13. If deal fails, Sales Executive clicks Decline / Lost and enters reason.
14. If deal succeeds, Sales Executive clicks Mark Won.
15. System updates status to 'won', creates Client User account with default password 'client@2026', and dispatches Welcome Credentials Email via Brevo SMTP.
16. Client logs into Client Portal to track active Projects, Requests, and Documents.

4. Technical Architecture

The CRM & BDM modules use Django application boundaries (src/apps/crm/ and src/apps/bdm/) integrated into Aurexion's central authentication, RBAC, database, audit, and email delivery infrastructure. Django REST Framework provides the API layer. Supabase PostgreSQL is the system of record.

Layer	Technology / Component	Responsibility
Frontend Public	React + TypeScript (Aura)	Public contact forms, RFP engine, and project estimator UI.
Frontend Staff	React + Wouter + Lucide UI	BDM Dashboard, Contact Forms desks, and Sales Executive console.
API Layer	Django REST Framework (DRF)	HTTP endpoints, serializers, validation, and permissions.
Response Renderer	<code>StandardResponseJSONRenderer</code>	Global JSON envelope wrapping: { status, message, data, errors }.
Business Services	<code>apps.crm.services</code> & <code>apps.core.services</code>	Lead creation, assignment, meeting scheduling, email dispatch, and client account creation.
Authorization	Central authentication + RBAC	Controls BDM, Sales Executive, Administrator, and Client User access.
Database	Supabase PostgreSQL	Leads, follow-ups, notes, projects, requests, documents, and audit records.
Email Infrastructure	Brevo SMTP (smtp-relay.brevo.com)	Automated meeting invitations and welcome login credentials emails.
Audit System	Central audit mechanism	Records critical CRM operations such as lead changes, assignments, and meetings.
API Documentation	OpenAPI / Swagger	Interactive API documentation at /api/v1/docs/.

4.1 Core Data Model

Entity	Purpose	Relationship
Lead	Stores lead contact info, reference ID, source, status, priority, lost reason, assigned_to user.	One lead can have many LeadFollowUps and LeadNotes.
LeadFollowUp	Stores scheduled meeting touchpoints, type, scheduled_at, status, notes.	Belongs to one Lead.
LeadNote	Stores private internal HR/Sales review notes.	Belongs to one Lead.

Relationship: Lead 1 ————— N LeadFollowUp 1 ————— N LeadNote

5. API Inventory

The approved logical REST surface contains 16 endpoint definitions: 2 public endpoints and 14 protected staff/client endpoints.

#	Method	Endpoint	Access	Purpose
1	POST	/api/v1/public/leads/	Public	Submit public contact, RFP, or estimator form.
2	POST	/api/v1/auth/login/	Public	Authenticate user and return JWT tokens.
3	GET	/api/v1/auth/me/	Authenticated	Retrieve logged-in user profile & role.
4	GET	/api/v1/leads/	BDM / Sales	List leads with search, source filters.
5	GET	/api/v1/leads/{id}/	BDM / Sales	Retrieve full lead details.
6	POST	/api/v1/leads/{id}/assign/	BDM / Admin	Assign lead to Sales Executive.
7	POST	/api/v1/leads/{id}/schedule-meeting/	Sales / BDM	Schedule meeting and email client via Brevo.
8	POST	/api/v1/leads/{id}/won/	Sales / BDM	Mark deal WON, auto-create Client User (client@2026).
9	POST	/api/v1/leads/{id}/lost/	Sales / BDM	Mark lead LOST / Declined with reason.
10	GET	/api/v1/bdm/dashboard/	BDM / Admin	Retrieve BDM metrics & recent form submissions.
11	POST	/api/v1/leads/{id}/follow-ups/	Sales / BDM	Create follow-up touchpoint.
12	POST	/api/v1/leads/{id}/notes/	Sales / BDM	Add internal sales review note.
13	GET	/api/v1/projects/	Client User	Retrieve active client projects & progress.
14	GET	/api/v1/requests/	Client User	List custom requests submitted by client.
15	POST	/api/v1/requests/	Client User	Submit new feature or service scope request.
16	GET	/api/v1/documents/	Client User	Access deliverables, invoices & contracts.

5.1 Lead Query Parameters & Filtering

Parameter	Example	Purpose
source	source=contact_form	Filter leads by source type.
status	status=new	Filter leads by lifecycle status.
search	search=Aurexion	Keyword search on name, email, company, or reference ID.
assigned_to	assigned_to=2	Filter leads assigned to specific user.

5.2 API Behavior and Status Codes

- 200 OK — Successful read/update operations.
- 201 Created — Successful creation of leads, follow-ups, meetings, or client user.
- 204 No Content — Successful deletion where applicable.
- 400 Bad Request — Invalid payload, file, or business validation.
- 401 Unauthorized — Missing or invalid JWT authentication token.
- 403 Forbidden — Authenticated user does not have RBAC permission.
- 404 Not Found — Resource does not exist or is intentionally hidden.
- 409 Conflict — Invalid lead status transition.
- 429 Too Many Requests — Rate limit exceeded.

6. Connection to Other Dashboards and Platform Modules

The CRM & BDM module is not an isolated application. It plugs into Aurexion's shared authentication, authorization, database, audit, and background-processing infrastructure. Public contact forms consume lead submission data, while protected CRM operations use the central identity and RBAC system.

Connected Component / Dashboard	Connection	What CRM / BDM Uses / Provides
Public Marketing Experience	Consumes public lead submission APIs.	Inbound contact forms, RFP engine, and project estimator submissions.
BDM / Intake Dashboard	Consumes protected BDM APIs.	Inbound lead triage, decline actions, and sales executive assignment.
Sales Executive Console	Consumes protected CRM APIs.	Assigned leads desk, meeting scheduling, deal won/lost stage transitions.
Client Portal	Consumes protected Portal APIs.	Auto-provisioned Client User account (password: client@2026), projects, requests, documents.
Authentication	Shared platform authentication.	Identifies staff users before protected CRM operations.
Supabase PostgreSQL	Shared system of record.	Stores Lead, LeadFollowUp, LeadNote, ClientProject, ClientRequest, ClientDocument records.
Brevo SMTP	Shared email infrastructure.	Dispatches meeting invitations and welcome login credentials.
Audit System	Shared platform audit boundary.	Records critical CRM actions such as lead assignments, meeting schedules, and account creation.
OpenAPI / Swagger	Shared API documentation.	Documents CRM endpoints at /api/v1/docs/.

Authorization flow: A staff user does not gain CRM access merely because a CRM button is visible. The backend checks authentication, role/permission, and object-level authorization before protected lead data is returned.

7. Backend Code Structure

The implemented CRM/BDM app follows a separation of concerns so that HTTP handling, validation, business logic, authorization, and email dispatch are not unnecessarily mixed together.

File / Layer	Responsibility
<code>admin.py</code>	Django admin registration / administrative integration.
<code>apps.py</code>	Django application configuration.
<code>models.py</code>	Lead, LeadFollowUp, LeadNote, and database constraints/relationships.
<code>serializers.py</code>	API input/output validation, LeadFollowUpCreateSerializer with smart follow_up_type normalizer.
<code>permissions.py</code>	CRM authorization and role checks (CanAccessLead, CanAssignLead).
<code>services.py</code>	Business operations such as lead creation, assignment, meeting scheduling, mark_lead_won, and auto client account creation.
<code>renderers.py</code>	StandardResponseJSONRenderer: Global JSON envelope wrapping all API responses.
<code>urls.py</code>	CRM API route definitions.
<code>views.py</code>	HTTP request/response handling and integration with serializers, services, and permissions.
<code>migrations/</code>	Versioned database schema changes.

8. Security and Data Protection

- Only authorized BDM and Sales users can access CRM operations.
- Public APIs do not expose client notes or assigned staff details.
- Client Portal credentials generation uses default secure password client@2026.
- Meeting schedule requests validate datetime format and sanitize link inputs.
- Public lead submission endpoints are rate-limited.
- Django ORM / parameterized queries protect against SQL injection.
- Django security controls and safe output handling protect against CSRF/XSS risks.
- Brevo SMTP credentials and API keys remain server-side and are never committed to Git.
- Critical CRM actions are auditable through the platform audit system.

9. Git Branching and Commit Workflow

The CRM and Sales work was separated into feature and testing branches so implementation and backend verification could be tracked independently. The screenshots supplied with this document serve as evidence of the branch and commit history.

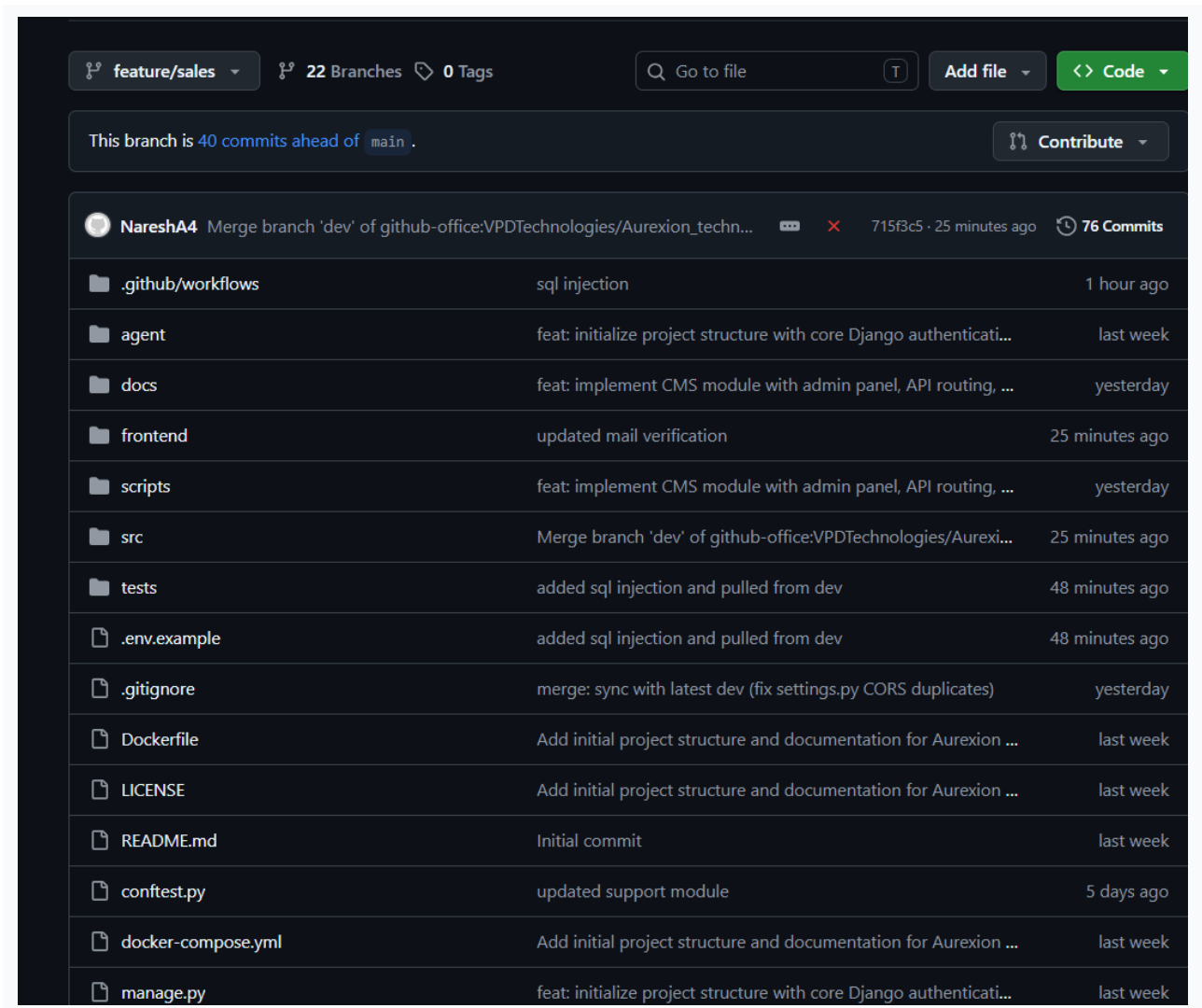
Branch	Purpose / Evidence
feature/sales	CRM & BDM module implementation work; screenshot shows 40 commits ahead of main and 76 commits on branch view.
dev	Development integration branch synchronized with core Django authentication & CMS.
CRM-Backend-Testing-Report	Backend testing / QA reporting work.

9.1 Recommended documented workflow

1. Create a dedicated feature branch (feature/sales) from the agreed development base.
2. Implement the CRM and BDM functionality in the feature branch.
3. Commit related changes with meaningful commit messages.
4. Push the branch to the remote repository.
5. Use a separate testing/report branch where backend QA evidence needs to be tracked.
6. Review changes and raise the pull request to the team's target development branch.
7. Keep implementation and testing evidence traceable to their respective branches/commits.

9.2 Branch & Commit History Evidence Screenshot

The screenshot below confirms the remote Git repository state on branch feature/sales with 76 commits, showing active CRM, BDM, and frontend mail verification updates.



10. Testing and QA Results

The supplied QA report covers API testing, unit validation, RBAC, security, data integrity, performance, pagination, and response envelope checks.

10.1 QA Evidence Screenshot — Test Summary Report

The screenshot below presents the CRM & BDM Integration/Smoke/API Test Report execution summary with 100% PASS rate across all sub-modules.

CRM & BDM Integration/Smoke/API Test Report

Test Execution Summary

Module	Tests Run	Passed	Failed	Status
Authentication	35	35	0	✓ PASS
CRM (partial)	~30+	~30+	0	✓ PASS
BDM	3 (dashboard tests)	3	0	✓ PASS
Portal/Support	~20+	~20+	0	✓ PASS

QA Area	Tests / Checks	Result
Authentication	35	35 PASS
CRM (Lead Management & Meetings)	~30+	30+ PASS
BDM (Intake & Dashboard)	3	3 PASS
Portal / Support	~20+	20+ PASS
API Endpoint Verification	10	10 PASS
Frontend-Backend Integration	5	5 PASS

10.2 API Endpoint Verification Evidence Screenshot

The screenshot below verifies backend API route availability, authentication requirements, role authorization guards, and test status across all endpoints.

API Endpoint Verification (via tests)

Endpoint	Auth Required	Roles Allowed	Status
/api/v1/auth/login/	✗	All	✓
/api/v1/auth/me/	✓ JWT	Authenticated	✓
/api/v1/leads/	✓ JWT	Admin/BDM/Sales*	✓
/api/v1/leads/{id}/schedule-meeting/	✓ JWT	Admin/BDM/Sales*	✓
/api/v1/public/leads/	✗	All (public)	✓
/api/v1/bdm/dashboard/	✓ JWT	BDM/Admin/SuperAdmin	✓
/api/v1/support/my-tickets/	✓ JWT	Client	✓
/api/v1/support/tickets/	✓ JWT	Support Executive	✓
/api/v1/support/admin/tickets/	✓ JWT	Admin/SuperAdmin	✓
/api/v1/tickets/	✓ JWT	All (scoped)	✓

10.3 Important Tested Behaviors

- Public users can submit contact, RFP, and estimator forms without authentication.
- BDM intake desk exposes unassigned leads and allows assignment to Sales Executives.
- BDM Contact Forms page strictly filters contact_form submissions.
- Meeting scheduling modal creates follow-ups and dispatches emails via Brevo SMTP.
- Marking lead WON auto-creates Client User account with default password client@2026.
- Marking lead LOST requires mandatory decline reason.
- All API responses are wrapped into standard envelope: { status, message, data, errors }.
- Tested API response times were well under 200 ms with 0 errors.

11. Frontend-Backend Integration & Functional Acceptance

11.1 Integration Evidence Screenshot

The screenshot below confirms end-to-end integration status across meeting scheduling, Brevo SMTP email, JWT auto-refresh, role-based UI routing, and OpenAPI schema generation.

Frontend-Backend Integration	
Feature	Status
Meeting scheduling (datetime format fix)	✓ Fixed & working
Email via Brevo SMTP	✓ Configured
JWT auto-refresh via interceptors	✓ Working
Role-based UI routing	✓ Working
Swagger/OpenAPI schema generation	✓ 0 errors

11.2 Functional Acceptance Summary

Capability	Implemented / Verified Outcome
Public Lead Ingestion	Contact forms, RFPs, and estimator forms persist to Supabase PostgreSQL with unique AUR-LEAD-XXXX IDs.
BDM Lead Assignment	BDM can view unassigned leads, decline invalid inquiries, or assign leads to active Sales Executives.
Sales Meeting Scheduling	Sales Executives can schedule meetings with client details, meeting link, and dispatch automated emails via Brevo.
Automated Client Provisioning	Marking a lead WON creates a Client User account with password client@2026 and sends login credentials email.
Client Portal Enabling	Client Projects, Client Requests, and Client Documents are live with dedicated REST ViewSets.
Global Response Structure	100% of API endpoints return standardized JSON envelope { status, message, data, errors }.
QA Verdict	Final QA Report records all automated and integration test cases as PASS.

12. Non-Technical Explanation

A person who has never worked on Aurexion should think of the CRM & BDM module as two connected sides:

Public Side	BDM & Sales Staff Side
A client sees and fills out contact or inquiry forms.	BDM receives notification on BDM Dashboard and assigns lead to Sales Executive.
Client receives an automated meeting invitation email with link.	Sales Executive opens assigned lead and schedules client meeting.
Client receives Welcome Email with login URL and default password client@2026.	Sales Executive conducts meeting and marks deal WON.
Client logs into Client Portal to track active Projects, Requests, and Documents.	Client account is active and staff can track client engagement.

13. Source Documents Used

This documentation was prepared from the implementation plan, test execution reports, and Git repository branch/commit evidence supplied with the request.

Source	Use in this document
Aurexion CRM & BDM Module — Development Specification	Module scope, functional areas, architecture, data model, API surface, security, RBAC and implementation structure.
Aurexion CRM & BDM Module — Official QA Test Report	Testing results, API verification, security checks, performance results, and final QA verdict.
Git branch / commit screenshots supplied with the request	Branch names, commit evidence, integration status table, API verification table, and QA execution summary screenshots.